

$$\textcircled{1} \quad f(x) = \sqrt{x} \quad x = 4, \quad \delta = 1$$

$$|x' - 4| \leq 1$$

a) δ_a exakt $\quad |f(x') - f(x)| \leq \delta_a |x' - x|$

b) δ_a ohne Schranke MWS

a) $|f(x') - f(x)| \leq \delta_a \quad \underline{|x' - x|} \quad \forall |x' - x| \leq \delta$

↑
beliebige Konstante

$$\frac{|f(x') - f(x)|}{|x' - x|} = \frac{|\sqrt{x'} - \sqrt{x}|}{|x' - x|}$$

$$= \frac{\cancel{|\sqrt{x'} - \sqrt{x}|}}{|\cancel{(\sqrt{x'} - \sqrt{x})}(\sqrt{x'} + \sqrt{x})|}$$

$$= \frac{1}{\sqrt{x'} + \sqrt{x}}$$

$$|x' - x| \leq \delta$$

$$x = 4, \quad \delta = 1$$

$$= \frac{1}{\sqrt{x'} + \sqrt{4}}$$

$$|x' - 4| \leq 1$$

$$x' \in [3, 5]$$

$$= \frac{1}{\sqrt{x'} + \sqrt{4}}$$

$$\xrightarrow{x'=3} \frac{1}{\sqrt{3} + 2} \approx \underline{\underline{0.268}}$$

$$b) \quad \underline{MWS} : \quad \Delta a \leq \underbrace{\max_{|x'-x| \leq \delta} |f'(x')|}$$

$$f(x) = \sqrt{x}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\max_{x' \in [3, 5]} |f'(x')| = \max_{x' \in [3, 5]} \frac{1}{2\sqrt{x'}}$$

$$= \frac{1}{2\sqrt{3}} \approx \underline{0.289} //$$